Sahil Khose

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Research Interests

world models · **VLMs** · **robustness**

I work toward **generalizable, physically-grounded embodied intelligence** – systems that perceive across modalities and act reliably under distribution shift. My current focus is **generative world models for robot learning** and **interpretable evaluation** to know when to trust them, building on prior work in **OOD robustness**, **synthetic data**, and **multimodal grounding**.

Education

University of California, Irvine, USA — 2026 – Present
Ph.D. in Computer Science — *GPA: 4.0/4.0*
Advisor: *Prof. Judy Hoffman*

Georgia Institute of Technology, Atlanta, USA — 2024 – 2026
Ph.D. in Computer Science — *GPA: 4.0/4.0*
Advisor: *Prof. Judy Hoffman*
Transferred to University of California, Irvine (advisor relocation)

Georgia Institute of Technology, Atlanta, USA — 2022 – 2024
M.S. in Computer Science (ML specialization) — *GPA: 4.0/4.0*
Thesis: *Improving Real-World Aerial Scene Understanding with a Synthetic Dataset* [ECCV 2024]
Committee: *Prof. Judy Hoffman (Advisor), Prof. Zsolt Kira, and Prof. Humphrey Shi*

Manipal Institute of Technology, Manipal, India — 2018 – 2022
B.Tech. in Computer and Communication Engineering — *GPA: 8.56/10.0*
Thesis: *Zero-Shot Domain Generalization: Unseen Classes in Unseen Domains*

Research Experience

University of California, Irvine, USA — 2026 – Present
Graduate Research Assistant at ***Hoffman AI Lab*** — *Advisor – Prof. Judy Hoffman*
- **WFM-Eval**: Developing VLM-based benchmarks for automatic evaluation of video world models in robotic manipulation.
- **VAM**: Generative video world models coupled with a flow-matching action expert for data-efficient robot policy learning.

Georgia Institute of Technology, Atlanta, USA — Jan 2023 – 2026
Graduate Research Assistant at ***Hoffman AI Lab*** — *Advisor – Prof. Judy Hoffman*
- **SPOT**: 3D Scene Graph Generation model for spatial and semantic reasoning in real-world robotics tasks. CVPR-W 2026
- **Multimodality**: Compact vision-audio LLM with MLP projectors and joint training, reduces cross-modal interference.
- **Syn-to-real:** Developed syn-to-real adaptation to raise off-road semantic segmentation performance.
- **SkyScenes:** Synthetic aerial benchmark that lifts model performance when transferring from syn-to-real. ECCV '24

Georgia Institute of Technology, Atlanta, USA — Spring 2023
Graduate Research Assistant at ***Neural Data Science Lab*** *(NerDS)* — *Advisor – Prof. Eva Dyer*
- **LatentDR**: a plug-and-play module to counter **diversity shift** without architecture changes. WACV '24

Relevant Research

WFM-Eval: Interpretable Error Diagnostics for Video World Models in Robotics

[Poster] Video World Models, [Poster] FMEA at ***CVPR 2026***
Under review at ***NeurIPS 2026***
Sahil Khose*, Mengqi Zhang, Prithvijit Chattopadhyay, Judy Hoffman*
A diagnostic framework for video world models in physical AI that identifies object hallucination as the dominant model-discriminative failure mode and exposes dataset-dependent ranking reversals across five recent models (Cosmos Predict2/2.5, Veo 3.1, HunyuanVideo 1.5, Wan2.2)

VAM: Generative Video Models for Robot Policy Learning
*Under review at **CoRL 2026***
Couples a video world-model expert with a flow-matching action expert through cross-attention bridge tokens grounded by affordance and depth, yielding data-efficient robot manipulation that outperforms prior video-action models on RoboCasa and transfers across LIBERO and real-world bimanual tasks.

SPOT: Structured Prompting with Object-centric Tokens for Open-World Scene Graphs
[Poster]** Multimodal Spatial Intelligence, **[Poster]** Visual Concepts at **CVPR 2026 Paper
*Mengqi Zhang, **Sahil Khose**, Fiona Ryan, Judy Hoffman*
Developed a scalable, 7B open-source VLM that unifies spatial and semantic scene graph generation, surpassing GPT-4o-based methods on both closed- and open-vocab 2D/3D SGG benchmarks.

Beyond Single Modalities: Lightweight Joint-Training for Vision + Audio Generalist LLMs *Preprint*
***Sahil Khose**, Manushree Vasu, Humphrey Shi, Judy Hoffman*
Designed a lightweight, jointly-trained 7B multimodal LLM that outperforms larger specialist and generalist models on vision and audio benchmarks by reducing cross-modal interference through simple MLP projectors.

SkyScenes: A Synthetic Dataset for Aerial Scene Understanding
*European Conference on Computer Vision **(ECCV) 2024*** *Paper* | *Dataset* | *GitHub*
Sahil Khose*, *Anisha Pal*, Aayushi Agarwal*, Deepanshi*, Judy Hoffman, Prithvijit Chattopadhyay*
Built a replayable CARLA pipeline that systematically varies viewpoint, weather, and lighting to surface domain-shift failure modes, then leveraged it to boost real-world aerial segmentation via syn-to-real transfer.

LatentDR: Improving Model Generalization With Sample-Aware Latent Degradation & Restoration
*Winter Conference on Applications of Computer Vision **(WACV) 2024*** *Paper*
*Ran Liu, **Sahil Khose**, Jingyun Xiao, Lakshmi Sathidevi, Keerthan Ramnath, Zsolt Kira, Eva L. Dyer*
A plug-and-play, sample-aware latent augmentation that lifts domain-generalization accuracy by up to 3 points on DomainBed and outperforms SoTA on medical and long-tail tasks.

Previous Research

INDICON 2023: Explainable Classification of Macular Degeneration Using Deep Learning *IEEE* | *Paper*
INDICON 2023: Fovea Segmentation Using Semi-Supervised Learning *IEEE* | *Paper*
NeurIPS-W 2022: Continual VQA for Disaster Response Systems *GitHub* | *Paper*
ICML-W 2022: An Efficient Modern Baseline for FloodNet VQA **BEST PAPER** *GitHub* | *Paper*
ACL-W 2022: Transformer based ensemble for emotion detection **ORAL** *GitHub* | *Paper*
NeurIPS-W 2021: A Studious Approach to Semi-Supervised Learning *GitHub* | *Paper*
NeurIPS-W 2021: XCI-Sketch: Colored Outlines & Sketches from Images **ORAL** *GitHub* | *Paper*
NeurIPS-W 2021: Semi-Supervised Aerial Classification & Segmentation **SPOTLIGHT** *GitHub* | *Paper*
NAACL-W 2021: BERT for Health Information Extraction from Social Media **TOP PERFORMER** *GitHub* | *Paper*

Professional Service

Conference Reviewer: **CVPR [2026, 2025]**, **NeurIPS [2026, 2025]**, **ECCV [2026, 2024]**, **CoRL 2026**
Workshop Reviewer: **CVPR-W 2026** (FMEA), **NeurIPS-W 2025** (MATH-AI), **CVPR-W 2025** (EMACS), **NeurIPS-W 2023** (ICBINB, DGM4H), **ICCV-W 2023** (WiCV), **NAACL-W 2021** (SMM4H)
Volunteer: **ICRA 2025** – Atlanta, GA, **NeurIPS 2022** – New Orleans, LA

ACHIEVEMENTS

Research Coverage

- **SkyScenes** (ECCV 2024) featured in Georgia Tech News, GT College of Computing, GT News Center, and Mirage News.

Research Awards

- **Best Paper Award** at the New In ML workshop. ICML 2022
- **Spotlight Paper** at the Tackling Climate Change with ML workshop. NeurIPS 2021
- **Top Performer Award** and special recognition on multi-task performance at the SMM4H workshop. NAACL 2021

Competitions

- **Project MANAS** stood **World Rank 1** at the **27th Intelligent Ground Vehicle Competition**. IGVC 2019
- **IGVC 2019 Awards:** Grand Award - 1st (Lescoe Cup), Interoperability - 1st, Design - 2nd, Cybersecurity - 3rd. IGVC 2019
- **Project MANAS** won the **Mahindra $1Million Challenge (top 13 out of 153 teams in India)**. 2019

PREVIOUS RESEARCH EXPERIENCE

Indian Institute of Science, Bangalore, India Jul 2021 – Jul 2022

AI Research Assistant at ***Artificial Intelligence and Robotics Lab*** *Advisors – Prof. S. Sundaram & Dr. Chandan Gautam*

- Bachelor's Thesis: Jointly addressing **domain shift** + **semantic shift** to recognize unseen classes in unseen domains.

Manipal Institute of Technology, Manipal, India Apr 2021 – Oct 2022

Medical AI Research Assistant *Advisor – Prof. Harish Kumar JR*

- Developed a medical diagnosis system for **fovea segmentation** using semi-supervised segmentation.
- Designed a **macular degeneration classification** system with interpretability for ophthalmology diagnosis.

Project MANAS – AI Robotics Research Team, MIT, Manipal, India Sep 2018 – May 2021

AI Perception Developer *GitLab* | *Website*

- **World Rank 1** at IGVC 2019 (UGV) & winner of the **Mahindra $1M Challenge** (out of 153 self-driving car teams).
- Implemented **Lane Detection, Speed Bump Detection, Driving Imitation System, Depth Map Generation**.

SELECTED PROJECTS

PR2. Domain Generalization: Tackling Diversity & Correlation Shifts *YouTube* | *GitHub* Fall 2022

- Unified RSC and VREx to jointly mitigate **diversity shift** + **spurious-correlation shift** by equalizing cross-domain risk and suppressing shortcut cues (e.g., dominant colors/edges).
- Established new SOTA on all six DomainBed datasets, with pronounced gains on color-biased gender-classification tasks.

PR1. Zero-Shot Domain Generalization: Unseen Classes in Unseen Domains *BTech Thesis* Spring 2022

- Developed a CLIP-powered Class-Normalization Zero-Shot Learning framework that jointly addresses **domain shift** + **semantic shift** on DomainNet, enabling one model to recognize unseen classes in unseen domains.
- Beat CuMix and DIN on all five held-out domains in $\sim$30 s/train run and proposed a realistic DGZSL evaluation protocol.

TEACHING EXPERIENCE

Graduate Teaching Assistant – CS 7647 Machine Learning with Limited Supervision *Course site* Fall 2023

- **Instructor: Prof. Judy Hoffman** | Guided 50 graduate students through state-of-the-art methods for visual learning with limited human supervision, mentoring 12 semester-long research projects from proposal to final evaluation.

TALKS

SkyScenes: Synthetic-to-Real Generalization for Aerial Imagery, NASA S2A2 Annual Meeting, Georgia Tech Jun 2023
An Efficient Modern Baseline for FloodNet VQA, New in ML @ ICML 2022 Jul 2022
Transformer-based Ensemble for Emotion Detection, WASSA @ ACL 2022 May 2022
XCI-Sketch: Extraction of Color Information from Images, New in ML @ NeurIPS 2021 Dec 2021
Semi-Supervised Classification & Segmentation on High-Res Aerial Images, CCAI @ NeurIPS 2021 Dec 2021
BERT Transformers for Extracting Health Information from Social Media, SMM4H @ NAACL 2021 Jun 2021